

MOHAMMED SAAQIB R

Computer Science & Artificial Intelligence Engineer | Healthcare & Full-Stack AI

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thiruk.vercel.app • hariharantraders.in

EDUCATION

SRM Institute of Science and Technology, Kattankulathur, Chennai — 2023 – 2027

B.Tech in Computer Science & Engineering (Artificial Intelligence and Machine Learning) — **CGPA: 7.26 / 10.0**

Chettinad Vidyashram, Chennai — Class XII (CBSE Science) — 2023 | **75.0%**

Chettinad Vidyashram, Chennai — Class X (CBSE) — 2021 | **70.0%**

TECHNICAL SKILLS

Programming Languages: Python, TypeScript, JavaScript, C, SQL (PostgreSQL), OOP Principles

Healthcare & Clinical AI: Ultrasound EF Video Tracking, Stethoscope Murmur AI, Drug Safety Safety Rules, DICOM/Medical Imaging

AI / ML & NLP: PyTorch, TensorFlow, HuggingFace Transformers, FAISS, Scikit-learn, OpenCV, Pandas, NumPy, RAG, CNNs

Full-Stack & Mobile: React Native, Expo, React.js, Next.js, Node.js, Express.js, WebSockets, HTML5, CSS3, REST APIs

Cloud, Databases & Tools: MongoDB, PostgreSQL, Firebase, AWS (S3, EC2), Docker, Git, GitHub, Vercel

APIs & Multimedia: Agora WebRTC, Twilio Video, SpeechRecognition, pyttsx3, gTTS/pydub

WORK & FREELANCE EXPERIENCE

Hariharan Traders — **Freelance Web Developer** [hariharantraders.in] — **Client Project | Chennai, India**

- Architected and launched the official commercial web platform for an established rice mill and wholesale export enterprise.
- Implemented an interactive showcase for Basmati, Parboiled, and Health rice lines with responsive UI/UX and mobile-optimized performance.
- Integrated direct customer inquiry and export quotation workflows, boosting digital reach and wholesale lead conversion by 35%.

Rocksphere Pvt Ltd — **Full-Stack Developer (Contract / Freelance)** — **Aug 2024 – Present | Chennai, India**

- Designed and managed corporate web applications, enhancing user interface responsiveness, loading speeds, and SEO visibility.
- Engineered an end-to-end custom online food ordering platform with real-time menu management, cart checkout, and kitchen order dispatch APIs.
- Built scalable backend services using Node.js and MongoDB for reliable transaction tracking, automated receipts, and operations management.

CLINICAL & HEALTHCARE AI PROJECTS

CardioSuite AI – Point-of-Care Clinical Workstation — *PyTorch, Ultrasound Video CV, Audio Signal Processing, Clinical AI* — **Cardiovascular AI**

- Architected a bedside point-of-care clinical workstation uniting ultrasound video ejection fraction (EF) tracking via deep learning computer vision for instantaneous cardiac function estimation.
- Engineered an acoustic signal processing and classification model for digital stethoscopes to detect and classify heart murmurs with high sensitivity.
- Integrated an organ-aware drug collision safety engine evaluating multidrug interactions and contraindications for comprehensive bedside cardiac risk evaluation.

VLinkScan – Mobile CT & MRI Medical Imaging Platform — *React Native, Node.js, WebGL, DICOM/Medical Imaging, AWS S3* — **Tele-Radiology**

- Developed a zero-footprint tele-radiology mobile application allowing patients and doctors to view, inspect, and analyze high-resolution CT and MRI scans directly on smartphones.
- Implemented multi-slice DICOM image parsing with windowing/leveling (Hounsfield units), multi-planar reconstruction, pan, zoom, and biomarker measurement tools.
- Engineered encrypted cloud pipelines on AWS S3 to ensure rapid, HIPAA-compliant medical scan sharing between imaging labs and consulting physicians.

Curelith – Multimodal AI Healthcare Assistant — *Python, HuggingFace, FAISS, OpenCV, React Native* — **Healthcare AI**

- Engineered an intelligent diagnostic assistant utilizing FAISS vector retrieval and HuggingFace LLMs indexed on verified medical databases (WHO, Mayo Clinic, NHS).
- Built an automated OCR/CV pipeline using OpenCV to analyze medical lab reports (PDF/images) and extract critical diagnostic indicators.
- Implemented speech-driven interaction with offline Speech Recognition, multilingual TTS, and diagnostic memory for continuous patient follow-ups.

FULL-STACK, ENTERPRISE & SYSTEMS PROJECTS

Thirukkural AI — *React, Next.js, NLP, LLM API, Vercel Serverless* [thiruk.vercel.app] **SRM Tamil Mandram | Live Product**

- Created an intelligent cultural knowledge portal providing semantic and natural language search across all 1,330 Thirukkural couplets.
- Integrated dual classical commentaries (Mu. Varadarajan, Solomon Pappaiah) and English translations with conversational life-advice matching.
- Deployed responsive serverless application on Vercel delivering sub-second search latency and high mobile usability.

Campus Placement Management System — *Node.js, Express.js, React, MongoDB, REST APIs* **Jeppiaar College | Enterprise Portal**

- Architected an end-to-end campus recruitment system automating student profile verification, drive eligibility checks, and resume submissions.
- Developed coordinator portals for drive scheduling, shortlisted candidate notifications, and round-wise interview tracking.
- Implemented administrative dashboards delivering real-time statistics on department-wise offers, CTC distributions, and recruitment metrics.

Core Banking Transaction System — *Python, Object-Oriented Programming (OOP), FinTech Architecture* **Systems Engineering**

- Engineered a reliable core banking ledger system using OOP design principles (modular encapsulation, data abstraction, and polymorphism).
- Implemented atomic transaction processing for account management, deposits, balance inquiries, withdrawals, and comprehensive audit logs.
- Created defensive validation mechanisms and automated exception handling to maintain database consistency and prevent concurrency anomalies.

Fuse – Real-Time Dating & Social Application — *React Native, Expo, Node.js, MongoDB, AWS S3, Agora SDK* **Mobile App Product**

- Built a Gen-Z dating application featuring real-time interest-based matchmaking algorithms and compliment-based chat unlocks.
- Integrated Agora and Twilio WebRTC SDKs for low-latency 1-on-1 video calls, in-call messaging, and secure AWS S3 media uploads.

LangBridge – Multilingual Chat SDK — *Node.js, WebSocket, React Native, Translation APIs* **Chat Technology**

- Designed a real-time translation chat overlay SDK enabling cross-language messaging across Discord and WhatsApp workflows.
- Maintained sub-second translation delivery across 10+ languages using bidirectional WebSocket streaming.

CERTIFICATIONS & FORMAL CREDENTIALS

- **AI in Healthcare Specialization** – Stanford Online / Coursera (Feb 2026) (5-Course Clinical AI Curriculum: Clinical Data, Machine Learning for Healthcare, Model Evaluations, and Capstone | Credential ID: `9B1HOKSZBPWV`)
- **Healthcare Management Assessment** – LearnTube.ai (July 2026) (Certified in healthcare administration protocols, operational metrics, and hospital management workflows | Certificate ID: `DJA-B-1-2762968-0`)
- **IBM Data Science Professional Certificate** – Coursera (July 2024) (Specialization covering Data Science Methodology, Python, SQL, Data Analysis, Machine Learning, and Applied Capstone | Credential: `5JCUDSE6ZK56`)
- **Certificate of Completion – C Programming** – SRMIST in collaboration with IEEE & Purdue University EPICS (Oct 2023) (Completed 50 hours of intensive programming and 100 problem-solving challenges on eLab | ID: `522284598929`)

HONORS, COMPETITIONS & CONFERENCES

- **Smart India Hackathon (SIH) 2025 (Internal Edition) – Finalist:** Presented an innovative circular bio-energy system converting agricultural waste and biogas into biomethane fuel for hybrid tractors.
- **Bug Buster's – Final Round (Sept 2024) – Finalist:** Qualified in the final round of the flagship coding and algorithmic debugging competition organized by CINTEL Student Association, SRMIST.
- **UNICEF All Political Party Meet for Child Welfare (May 2025) – Invited Delegate:** Presented a large-scale AI healthcare architecture for pediatric mental health screening and youth emotional well-being at Anna University.
- **8th International Conference on Nanoscience & Nanotechnology (ICONN-2025) – Volunteer:** Assisted in event coordination and technical sessions alongside international researchers and academic partners.

LANGUAGES

English (Professional Working Proficiency) • **Tamil** (Native Proficiency) • **Hindi** (Elementary Proficiency)